

Product Requirements Document: Website Checker

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1. Introduction & Vision

1.1. Overview

The Website Checker is a full-stack web application designed to automate the process of monitoring website health and integrity. It provides a centralized dashboard to manage a list of URLs, run on-demand or scheduled checks, and receive detailed reports on the status of each site.

1.2. Problem Statement

Manually checking a portfolio of websites for issues like downtime, SSL errors, database connection problems, or unwanted content is inefficient, repetitive, and prone to human error. Site administrators need an automated tool to perform these checks reliably, maintain a historical log, and be alerted to problems in a timely manner.

1.3. Goals & Objectives

- **Automate:** Eliminate the manual effort of routine website health checks.
- **Centralize:** Provide a single dashboard to view the status of all monitored websites.
- **Inform:** Deliver automated daily email reports summarizing the latest check results.
- **Diagnose:** Capture visual screenshots and specific error messages to help quickly identify problems.

2. User Persona

- **Title:** Site Administrator / Freelance Developer
- **Description:** An individual who manages multiple websites for clients or personal projects.
- **Needs:**
 - An efficient way to monitor the uptime and health of all sites.
 - A historical record of site performance and issues.
 - Proactive notifications about problems before clients notice them.
- **Pain Points:**
 - Time wasted on repetitive manual checks.
 - Anxiety about a client's site going down without their knowledge.

- Difficulty tracking the intermittent nature of certain website errors.

3. Features & Capabilities

3.1. Core Monitoring & Checks

- **Uptime & Status Check:** Verifies that the website is online and returns a successful HTTP status code (2xx). Flags any client (4xx) or server (5xx) errors.
- **Response Time (Latency):** Measures the time it takes for a website to respond to a request, tracking performance over time.
- **SSL Certificate Validation:** Automatically checks that the SSL certificate is valid, trusted, and not nearing its expiration date.
- **Content & Keyword Analysis:**
 - **Required Text:** Checks for the *presence* of specific text (e.g., a company name, a copyright notice) to ensure the correct page is loading.
 - **Unwanted Text:** Scans the page for negative keywords indicating a problem, such as database errors or adult content.
- **Broken Link Checker:** Crawls the homepage (or specified pages) to find and report any broken internal or external links (404 errors).
- **Screenshot Capture:** Takes a full-page screenshot with every check, providing a visual record of the site's appearance and proof of any visual errors.
- **Basic SEO Health Check:** Scans for the presence, length, and content of essential on-page SEO elements like <title> tags and <meta name="description">.

3.2. Dashboard & User Interface

- **Centralized Dashboard:** A single-pane view showing the latest status of all monitored websites with key metrics.
- **Advanced URL Management:**
 - **CRUD Operations:** The ability to easily add, edit, and delete URLs.
 - **Grouping & Tagging:** Organize monitored websites by client, project, or priority for easier management.
 - **Bulk Import/Export:** Add or download a list of URLs using a CSV file.
- **On-Demand Testing:** Trigger an immediate check for a single site or all sites with the click of a button.
- **Historical Data Visualization:** A detailed view for each website showing a historical graph of its uptime and response time over the last 7, 30, or 90 days.

3.3. Reporting & Notifications

- **Scheduled Summary Reports:** Automatically generate and email daily or weekly summary reports to a list of specified recipients.
- **Instant Failure Alerts:** Send an immediate notification the moment a check fails, allowing for rapid response to critical issues.
- **Multi-Channel Notifications:** Integrate with various platforms to deliver alerts where

users are most active (Email, Slack, SMS, etc.).

- **Configurable Alerting:** Set custom alert thresholds to reduce notification fatigue (e.g., alert only after 2 consecutive failures).

3.4. User & Account Management

- **Secure User Authentication:** Allow users to sign up and log in to manage their own private list of monitored websites.
- **Team & Workspace Support:** Enable an account owner to invite team members to view and manage a shared set of websites.

4. Technical Architecture

Component	Technology/Service
Frontend	React (Vite)
Backend	Node.js with Express.js
Deployment	Frontend on Vercel, Backend on Render (Docker)
Database	Supabase (PostgreSQL)
File Storage	Supabase Storage
Checking Engine	Puppeteer (Headless Chromium)
Scheduling	node-cron
Email Service	Nodemailer with SendGrid Transport

5. Roadmap & Future Considerations

Version 1.1

- Implement a detailed historical view for each monitored URL.
- Add user authentication and accounts.
- Allow users to set custom check frequencies per URL.

Version 1.2

- Introduce advanced checks for SEO elements and broken links.
- Integrate additional notification channels like Slack.
- Develop public-facing status pages.